

Pre-registration form

Chromatic discrimination in the lower and upper visual fields

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Purpose

Natural visual environments contain strong asymmetries in the light reaching the eye from the upper and lower visual fields. Previous work measuring directional structures of natural lighting environments has shown pronounced elevation-dependent differences in both intensity and spectral composition. On clear days, the upper visual field is often dominated by bright sunlight and blue skylight, whereas light from the lower visual field is dominated by inter-reflection from the ground and surrounding surfaces. As a result, the overall intensity of light can differ substantially across elevation, often by approximately an order of magnitude, and the spectral composition of lower-field illumination can vary depending on the reflectance properties of nearby objects and the ground. Although this asymmetry is attenuated under overcast skies and in indoor environments, it remains a robust and ecologically relevant feature of the visual input.

If such upper–lower asymmetries are a persistent feature of natural visual input across the lifespan, then photoreceptors and post-receptoral mechanisms serving different retinotopic regions are chronically exposed to different distributions of light; that is, different visual field locations experience a different spectral diet. This raises the possibility that the visual system is calibrated in a spatially localized manner, such that mechanisms at different visual-field locations are adapted to the input statistics characteristic of those locations. For instance, the upper visual field may trade off fine chromatic discriminability in order to accommodate the wide dynamic range of light it typically receives, whereas the lower visual field may favour finer sensitivity over a narrower range.

The present project addresses this question by comprehensively comparing the sensitivity as measured by visual discrimination in 3D colour space across the upper and lower visual fields.

Plan

This preregistration document describes an initial methods refinement experiment in which we will make an iterative set of measurements of colour discrimination measurements around D65 at an upper and lower visual field location, for three experienced participants. For each visual-field location we will estimate a three-dimensional colour discrimination ellipsoid centred on D65 using the recently developed efficient threshold estimation method called Wishart Process Psychophysical Model (WPPM; Hong et al., 2025, eLife).

Our goal for this initial experiment is to gain experience with the WPPM as it will be applied here, and with how to use eye tracking measurements to monitor fixation and inform trial exclusion. We view it as reasonably likely that we will decide to collect additional data beyond that described here from the initial three experienced participants.

Hypotheses

We have the following hypotheses.

Hypothesis 1. The three-dimensional chromatic discrimination ellipsoid around D65 will differ systematically between the upper and lower visual fields, in its size, and/or orientation.

Hypothesis 2. The structure of these threshold differences will correspond to differences in natural scene statistics, with the direction and magnitude of perceptual sensitivity differences reflecting the upper–lower asymmetries observed in natural chromatic input.

Methods

Apparatus

Stimuli will be presented on a 32-inch LCD monitor (Display++, 1024 × 1024 pixels, 14-bit per gun, frame rate 120 Hz; Cambridge Research Systems, Rochester, Kent, UK), enabling fine control of chromatic and luminance modulation.

Participants will view the display binocularly from a fixed distance of 59 cm. The head will be stabilised using a chin rest, with the forehead also supported, so that head position is held constant throughout the experiment.

Eye movements will be recorded throughout the experiment using an EyeLink 1000 eye tracker (1,000 Hz sampling rate, 0.25–0.50° typical accuracy, 0.01° RMS resolution; SR Research, Ottawa, Ontario, Canada). In this experiment, we will record eye movements and determine post-hoc criteria for excluding trials based on fixation failures, as at present we are unsure of typical magnitude and rate of such failures.

Spectral calibration of the display primaries was performed with a spectroradiometer (SpectraScan PR-670, spectral range 380–780 nm, ± 1 nm wavelength accuracy; Photo Research, Chatsworth, CA, USA), and gamma correction was carried out using a colorimeter (ColorCAL MKII; Cambridge Research Systems, Rochester, Kent, UK). From these measurements we constructed the inverse-gamma look-up tables and the linear transformation between device (RGB) coordinates and the cone-based / colorimetric stimulus coordinates (LMS / CIE XYZ).

Stimuli

The stimuli to be discriminated are presented against a spatially uniform background with the chromaticity of D65 and luminance of 50.9 cd/m². These stimuli will consist of Gaussian blobs, chosen to avoid sharp chromatic and luminance edges. Each blob

has a spatial Gaussian profile with a standard deviation of 0.5° , giving a visible diameter of approximately 2° . The stimulus is windowed in time by a temporal Gaussian with a standard deviation of 0.35 s, so that it ramps on and off gradually. The total stimulus presentation duration is 1 s.

On each trial, three Gaussian blobs are presented simultaneously on an iso-eccentric circle at 10° eccentricity, in either the upper visual field (centred 10° above fixation) or the lower visual field (centred 10° below fixation), with adjacent locations separated by 15° of polar (arc) angle (Figure 1). The central location always contains the reference. The two flanking locations, to its left and right, contain one patch that is identical to the reference and one that is chromatically displaced from it — the target — by a magnitude and direction specified by the adaptive psychophysics tool called AEPsych. The observer's task is to report which side, left or right, contains the patch that differs from the central reference (a spatial two-alternative forced choice). A black fixation cross is present at the centre of the display throughout, and participants maintain fixation on it; all stimulus locations are defined relative to this cross.

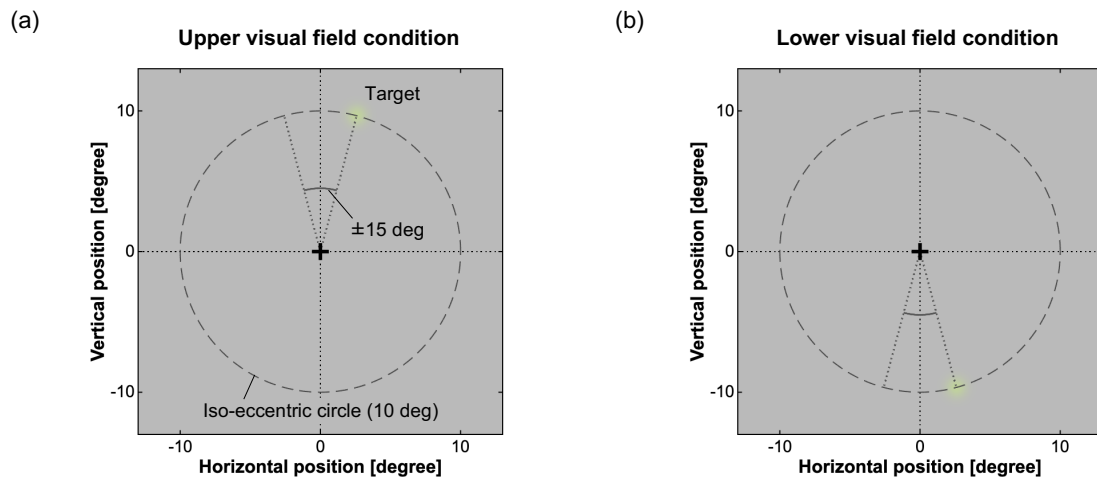


Figure 1. Schematic of the stimulus configuration for Experiment 1, shown for (a) the upper-field condition and (b) the lower-field condition. On each trial the stimuli lie on an iso-eccentric circle at 10° eccentricity (dashed circle): a reference at D65 is presented at the central location on the vertical meridian (not drawn, because it is identical to the D65 background), and two stimuli are presented at $\pm 15^\circ$ of polar (arc) angle from the meridian along the circle. One of these is the target stimulus and the other is the distractor stimulus. The participant reports whether the target appeared on the left or the right (spatial two-alternative forced choice).

In this experiment, the reference is D65 and is therefore identical to the background, so the central reference and the matching flanker are not visible; in practice the observer sees only the displaced target and reports its side. The task is framed in this way because it generalises directly to subsequent experiments, in which the reference will be varied across different locations in 3D colour space.

Stimuli are specified in the model space of the WPPM. This space is essentially the monitor's linearized RGB gamut cube, with each axis parameterized from -1 to 1 instead of the typical 0 to 1. The parameterization is chosen for technical convenience

of fitting the WPPM (see Hong et al, 2025, eLife). Stimuli will be sampled in three dimensions around the D65 reference, limited to the central range of -0.25 to 0.25 based on pilot data. This range may be refined as we proceed.

Because this space is related by a linear transformation to CIE XYZ and to cone-based LMS coordinates, the resulting thresholds can be expressed as a three-dimensional ellipsoid in standard colour coordinates.

Procedure

On each trial the participant fixates the central cross on the stationary D65 background, and the stimuli are presented for 1 s, after which the participant reports the side (left or right) of the target (Figure 1). Outside these brief presentation intervals the display remains at the D65 background, so that observers stay adapted to D65 throughout. Eye position is recorded on every trial and will be used to exclude trials with excessive eye movements, with criteria to be determined in this initial experiment as we gain understanding of typical fixation precision. This may involve some pragmatic tradeoffs between minimizing effects of fixation error and maximizing number of trials available for WPPM fitting.

Stimuli are placed adaptively by AEPsych: each block begins with a set of Sobol' quasi-random trials and then proceeds to model-guided placement. The range over which the initial Sobol samples are drawn was determined based on one session of pilot data from the upper visual field location.

We will nominally collect 500 AEPsych (Sobol' + model guided) trials per visual-field location, for a total of 1,000 trials per participant. In addition, fallback trials generated by Sobol' sampling will be inserted when the model guided trial placement times out. This trial number was chosen based on simulations with a computational observer that made trial-by-trial decisions from the CIEDE2000 difference between the target and the reference; in simulation, 500 trials per location were generally sufficient to recover the simulated ground truth, although not perfectly. The initial data set with human participants will allow us to more precisely adjust trial placement and trial number parameters, based on bootstrapping to estimate measurement variability. The magnitude of the Sobol' trial colour distance between reference and comparison as well as the number of Sobol' versus model guided trials will be reported with the data. These may be refined across the three participants as we proceed.

The upper- and lower-field measurements are run in separate blocks, because interleaving the two locations within a block would make the task harder and prevent chromatic sensitivity from being measured as accurately. Each nominal 500-trial block is completed in a single, self-paced run, with an opportunity to rest every 100 trials. Before the main experiment, participants complete practice trials, which are excluded from the analyses.

As noted above, because AEPsych sometimes is unable to determine the next model-guided trial before the expected time of the next trial, we will slot in Sobol' generated fallback trials when this happens.

Participant eligibility and exclusion criteria

We will recruit adult participants with normal or corrected-to-normal visual acuity and normal colour vision. Participants will be screened using standard tests of colour vision (e.g. Hardy-Rand-Rittler plate) prior to the main experiment. All participants will be 18 years of age or older and will provide written informed consent under an approved ethics protocol. We aim to test 3 experienced participants for Experiment 1. These participants are all experienced psychophysical observers. This study has been reviewed and approved by the Central University Research Ethics Committee (CUREC) at University of Oxford.

Participants will be compensated at the approved rate of £12 per hour.

If participants complete only part of the experiment, their data may still be included provided that sufficient data are available for stable estimation of the thresholds.

Data analysis

Discrimination-ellipsoid estimation

Behavioural data will be analysed with the WPPM to estimate the three-dimensional colour discrimination ellipsoid around D65 separately for the upper and lower visual fields, under the assumption that sensitivity varies smoothly across colour space.

From these fits we extract summary measures of each ellipsoid, including its volume, the lengths of its principal axes, and its orientation in colour space.

We will use bootstrapping to estimate 95% confidence intervals around the measured discrimination ellipsoids, based on 120 bootstrap iterations.

One thing we will look at closely is whether the trial placement appears to constrain the full surface of the estimated ellipsoid, or whether there are directions where too few trials appear to have been placed. It is possible that we will use the fit from the first session to design trial placement for additional sessions, by placing trials near the initial estimated threshold surface.

Comparison between upper and lower visual fields

The primary comparison is the difference in the discrimination ellipsoid between the upper and lower visual-field locations: whether it is systematically larger, smaller, stretched, compressed, or reoriented in one field relative to the other, relative to the estimated 95% confidence intervals.

We will also test whether this upper–lower difference mirrors the differences in natural-scene chromatic statistics.

Statistical analysis

In addition to visual comparisons of the measured ellipsoids, the main comparisons will use within-subject statistical analyses appropriate for repeated-measures data. These may include paired-samples tests and/or linear mixed-effects models with visual-field location as a fixed effect and participant as a random effect.

We report effect sizes and confidence intervals alongside inferential statistics.